RAFEEQ SmartWheel

6 SLIDES · FINAL

Electrical Subsystem — Slide Implementation Brief

For: Presentation Developer / HTML Engineer

CIE 598/599 · Team 06 · Zewail City of Science and Technology · 2025/2026

Content sourced from thesis §4.3.2 (Power Architecture) · Verified against all data tables

HOW TO USE THIS DOCUMENT

This brief covers the 6 Electrical subsystem slides in the HTML presentation. All numbers are verified against the thesis (§4.3.2). One error corrected from the draft narrative: the motors use **8-inch** hub wheels for drive, mounted on 10-inch pneumatic tyres (§4.3.3.1). The battery comparison table (Section 7) is new — not in the original brief.

Slides covered	6 slides: ELEC-1 (Hook) · ELEC-2 (Architecture) · ELEC-3 (Power Distribution) · ELEC-4 (Motors) · ELEC-5 (Battery Runtime) · ELEC-6 (Safety)
Insertion point	After mechanical slides (slides 10–13). Before Control section (slides ~16–20). Electrical becomes slides 14–19.
Thesis source	§4.3.1 Overview · §4.3.2 Power Architecture · §4.3.3 STM32 Driver · §4.3.2.3 Safety
Brand colours	Navy #0D1F3C · Gold #C9992A · Blue #3B7DE8 · Green #27AE60 · Red #C94040
Key correction	Draft said "10-inch hub motors" — correct phrase is "350W BLDC hub motors, 8-inch wheel diameter" (§4.3.3.1)
New addition	Section 7: Battery Before vs After comparison table — thesis Tables 4.4 vs 4.6. Use in ELEC-5 slide.

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SECTION 1

Overview & Insertion Point

Section Story Arc

"The mechanical chassis is now built. What makes it intelligent is the electrical backbone — engineered around one core philosophy: Total Power Isolation."

#	Slide	Story Beat	Thesis Ref	Time
ELEC-1	The Hook	Transition from mechanical. Core philosophy: Total Power Isolation.	§4.3.1	~30s
ELEC-2	Three-Tier Architecture	3 tiers: RPi5 (think) → Arduino (control) → STM32 (actuate). Fig 4.9.	§4.3.1 Fig 4.9	~1 min
ELEC-3	Power Distribution	Two isolated battery domains. 36V traction + 12V electronics. 3 buck rails.	§4.3.2 Fig 4.10	~1 min
ELEC-4	Motors & Power Draw	FOC explained. 3 current profiles. Why FOC = smooth patient ride.	§4.3.3.1–4.3.3.2	~1 min
ELEC-5	Battery Runtime + Comparison	Single-pack vs dual-pack table. Runtime doubled. Voltage sag eliminated. Production case.	§4.3.2.4 Tables 4.4/4.6	~1.5 min
ELEC-6	4-Layer Safety Net	BMS → Fuses → Watchdog → E-Stop. Defence-in-depth. Fig 4.12.	§4.3.2.3 Fig 4.12	~1 min

Insertion Point in rafeeq_final.html

Insert after	Last mechanical slide (slide-13 in current numbering, after Mechanical section)
New IDs	id="slide-14" through id="slide-19" · hooks window._slide14 through window._slide19
Total after insert	29 slides (5 control + 6 electrical + original 24 – 6 mechanical already added = check current total)
Note	Coordinate with whoever built the mechanical and control slides — all insertion points must be sequenced together

SECTION 2

Slide ELEC-1 — The Hook: From Structure to Superpower

Dark transition slide. ~30 seconds. No interaction needed.

ELEC-1	The Hook — From Structure to Superpower	DARK · TRANSITION

Purpose

Bridges the mechanical section to the electrical section. Sets up the core philosophy. High impact, minimal text. Dark navy background, one headline, one powerful supporting line.

Exact Slide Content

Element	Exact Text	Style
Section chip	05	Large ghost number 80px, gold 12% opacity
Eyebrow	Methodology & Implementation · Electrical Subsystem	Gold, 9px, all-caps, letter-spaced
Headline line 1	Total Power	White, 48–80px, weight 900
Headline line 2 (gold)	Isolation.	Gold #C9992A, same size/weight
Gold bar	—	3px × 80px gold bar, expands on enter
Body line	We designed it so that the raw muscle powering the wheels never interferes with the intelligence running the AI.	White 50% opacity, 15px, max-width 560px
4 Pills	36V Traction · 12V Electronics · Galvanic Isolation · Defence-in-Depth	Same pill style as CTRL-1

Animation Sequence

Step	Element	Animation	Delay
1	Ghost chip "05"	translateX(-20px)→0, opacity 0→1	0ms
2	Eyebrow	opacity 0→1	150ms
3	Headline (both lines)	translateY(20px)→0, opacity 0→1	250ms
4	Gold bar	width 0→80px	500ms
5	Body paragraph	opacity 0→1	620ms
6	All 4 pills together	translateY(14px)→0, opacity 0→1	780ms

SECTION 3

Slide ELEC-2 — Three-Tier Architecture

Same tier diagram as CTRL-2 but presented from electrical perspective. Thesis §4.3.1 Fig 4.9.

ELEC-2	Three-Tier Architecture	DARK · ANIMATED

NOTE: This slide covers the same three-tier hierarchy as Control slide CTRL-2. In the final presentation, these may be merged into one slide or ELEC-2 may be skipped if CTRL-2 already covers it. Check with the presenter. If kept: frame it from an ELECTRICAL perspective (power flow, not data flow).

Exact Tier Content — Electrical Perspective

Tier	Hardware	Electrical Role	Power Source
Tier 1 — High-Level Compute	Raspberry Pi 5 (8GB)	Runs SLAM, Nav2, ASR, Firebase IoT. Ubuntu 24.04 LTS, ROS2 Jazzy.	Rail 1: XL4016 → 5.1V/5A from 12V pack
Tier 2 — Real-Time Control	Arduino Mega 2560	50Hz deterministic loop. Kinematics, watchdog, serial bridge.	Rail 3: LM2596 → 5V/1A from 12V pack
Tier 3 — Actuation	STM32F103RCT6 + 2×350W BLDC	FOC 20kHz commutation. Direct motor drive.	36V Li-ion pack (VICTa 10S2P) via XT60

Inter-Tier Communication (show as wire labels on diagram)

From	To	Protocol	Rate	Electrical Interface
RPi5 (Tier 1)	Arduino (Tier 2)	V,W<ω>\n ASCII	20 Hz	USB-CDC /dev/ttyACM0 · 5V logic
Arduino (Tier 2)	STM32 (Tier 3)	Binary frame 0xABCD · 8 bytes	50 Hz	UART1 Serial1 pins 18/19 · 5V→3.3V
STM32 (Tier 3)	Arduino (Tier 2)	Feedback 18 bytes · Hall+batV+temp	50 Hz	UART feedback · 3.3V logic

Animation: same as CTRL-2. Each tier slides in from left, staggered 180ms. Feedback badge pulses green at bottom.

SECTION 4

Slide ELEC-3 — Power Distribution (Split-Battery Network)

The two isolated domains + three-rail electronics supply. Thesis §4.3.2 Figs 4.10/4.11.

ELEC-3	Power Distribution — Split-Battery Network	DARK · DIAGRAM

Core Engineering Principle (use as headline)

"20kHz motor PWM switching generates conducted noise up to 15A peak per phase. If the electronics shared the same battery, this noise would corrupt serial comms and sensor data. Galvanic isolation via separate packs and dedicated DC–DC rails eliminates this coupling path entirely." (§4.3.2)

Domain 1 — Motor Traction Battery (36V, LEFT side of diagram)

Battery	VICTa10S2PND2200P01 · Li-ion · 10S2P Samsung cylindrical cells
Nominal voltage	36V (cell range: 30V depleted → 42V full)
Capacity	4.4 Ah per pack · 158.4 Wh per pack
Connector	XT60 · 12 AWG wiring · 30A inline automotive blade fuse per positive line
Connected to	STM32 FOC motor driver → 2×350W BLDC hub motors
Fuse sizing	IEC 60364-5-52: I_derated = 0.8×41A = 32.8A → select 30A (below derated limit, above 15A continuous rating)
BMS protection	Per-cell: 4.2V max / 2.5V min / 60°C over-temp disconnect

Domain 2 — Electronics Rail (12V, separate pack)

Battery	12V Li-ion pack · 3Ah · Nominal 12V
Master fuse	10A upstream of all buck converters
Runtime	12×3/25 ≈ 1.44h on electronics load (§4.3.2.2 Eq.4.16)
Architecture	3 galvanically isolated DC–DC buck converter rails (Table 4.5)

Three Electronics Rails — Exact Spec (Table 4.5)

Rail	Converter	V_out	I_max	Load Connected	Why This Converter
Rail 1	XL4016	5.1V	5A	Raspberry Pi 5	RPi5 draws up to 5A peak under heavy CPU. LM2596 (3A max) would cause undervoltage + CPU throttling. XL4016 = 94% efficient. P_loss ≈ 0.97W (no heatsink needed).
Rail 2	LM2596	5.0V	3A	USB hub · RPLidar A1M8 · RGB camera	Moderate load. LM2596 sufficient.
Rail 3	LM2596	5.0V	1A	Arduino Mega 2560 · MPU-6050 · auxiliary sensors	Low load. Each rail independently fused for fault isolation.

Slide visual: Two-column diagram. LEFT = 36V domain (red/orange). RIGHT = 12V domain (blue/teal). Arrow flows show power direction. Reference Figure 4.10 (wiring photo) and Figure 4.11 (block diagram) from thesis.

SECTION 5

Slide ELEC-4 — The Muscles: Motors & Real-World Power Draw

FOC explained + 3 current profiles. Thesis §4.3.3.1–4.3.3.2.

ELEC-4 | **The Muscles — Motors & Real-World Power Draw** | **DARK · DATA**

★ Motor Specification (CORRECTED from draft narrative)

The draft brief said "10-inch hub motors". The correct description from §4.3.3.1 is: "two 350W BLDC hub motors, 8-inch wheel diameter." The BOM (Appendix B) confirms: "BLDC Hub Motors, 350W, 8-inch" — 8-inch is the wheel diameter. Appendix A mentions "10-inch pneumatic wheels" referring to the outer tyre fitted on the hub — but the motor component itself is the 8-inch hub. Use "8-inch" throughout.

Motors	2× BLDC hub motors · 350W each · 8-inch wheel diameter (BOM Appendix B; §4.3.3.1)
Driver board	STM32F103RCT6 hoverboard motherboard · reflashed with EFeru FOC firmware via ST-Link V2
Commutation	Sinusoidal FOC at 20kHz · entirely internal to STM32 · zero burden on Arduino
Why FOC?	Trapezoidal (6-step) commutation causes torque ripple → vibration perceptible by a seated patient. FOC maintains stator current orthogonal to rotor flux → zero ripple → silent ride.
Control mode	VLT_MODE (voltage-mode, not speed-mode) — cmd=0 freewheels, allows manual pushing
N_MOT_MAX	90 RPM → $v_{max} = 90 \times 2\pi/60 \times 0.130 \approx 1.22$ m/s (ISO 7176-14 indoor speed limit)

Three Current Profiles — Exact Numbers (Table 4.4 source data)

These are TOTAL current draw from both motors combined. Each motor draws half (1A / 1.75A / 4A per motor from §4.3.2.1).

Usage Profile	I_avg Total (A)	P_avg (W)	Description	Colour code
Light indoor	2.0A	72W	Slow cruising, clear corridor. Both motors @1A each.	Green #27AE60
Normal indoor	3.5A	126W	Mixed use: corridors, gradual turns. Both motors @1.75A each.	Gold #C9992A
Heavy load	8.0A	288W	Ramp traversal, sharp combined turns. Both motors @4A each.	Red #C94040

Slide visual: Three cards (green/gold/red) with icon, current value, and one-line description. FOC explanation: show waveform comparison — trapezoidal (jagged steps) vs sinusoidal (smooth wave). Source: §4.3.3.2.

SECTION 6

Slide ELEC-5 — Battery Cycle: Runtime & Charging

Dual-pack modification story + runtime table. Thesis §4.3.2.1 Tables 4.4/4.6.

ELEC-5	Battery Cycle — Runtime & Charging	DARK · TABLE + STORY

The Story Arc on This Slide

This slide tells a problem → engineering solution → result story. Three acts: (1) We discovered a 4–6V sag. (2) We connected two packs in parallel via XT60 Y-harness. (3) Runtime doubled, sag eliminated. Then show the numbers.

■ Problem Observed	■ Engineering Fix
During seated integration testing: combined forward + rotation manoeuvres caused a 4–6V sag on the 36V bus. STM32 triggered BAT_DEAD cutoff. Wheelchair stopped mid-turn. (Primary cause: DEFAULT_RATE=800 firmware setting. §4.3.5.1)	Secondary hardware fix: two VICTa packs connected in parallel via XT60 Y-harness, 12AWG wire, 30A inline fuse per line. $R_{eff} = R_{pack}/2 \rightarrow \Delta V_{sag}$ halved. Pre-condition: both packs within 0.2V before closing Y-harness. Result: $E_{total} = 2 \times 158.4 = 316.8$Wh. Runtime doubled. (§4.3.2.4)

Runtime Numbers — Dual Pack (Table 4.6) — Use on Slide

Profile	I_avg	P_avg	Runtime (h)	Runtime (min)	Daily Life Translation
■ Light indoor	2.0A	72W	4.40h	264 min	Full half-day of light indoor cruising
■ Normal mixed	3.5A	126W	2.51h	151 min	Typical daily use — more than sufficient
■ Heavy load	8.0A	288W	1.10h	66 min	Worst-case: constant ramp climbing

Charging Times — Use as Bottom Strip on Slide

Configuration	Charger Setup	Time (h)	Thesis Eq.
Dual pack	2× 2A standard (one per pack)	2.4h	Eq. 4.26
Dual pack	2× 4A fast (one per pack)	1.2h	Eq. 4.27
Dual pack	1× 2A sequential	4.9h	Eq. 4.28

■■ Important note for presenter (§4.3.2.2): After the dual motor-pack modification, the electronics pack (1.44h on the 12V/3Ah pack) becomes the binding constraint on total system uptime. Solution deferred to future revision: upgrade to 6Ah electronics pack → $t_{elec} = 12 \times 6/25 \approx 2.88$h. Do NOT say this on the slide — it is a known limitation. Mention only if the committee asks.

SECTION 7

★ Battery Before vs After Comparison (NEW)

Full comparison of single-pack vs dual-pack. Use as the centrepiece visual of ELEC-5.

Why This Matters for the Production Argument

This table is the engineering evidence that the dual-pack modification is not just a fix — it is a design decision that makes RAFEEQ viable as a production assistive device. It addresses both the voltage stability problem AND the runtime requirement for daily use.

The Full Before vs After Comparison Table

Use this as a full-width table visual in ELEC-5. Colour the AFTER column gold/green.

Parameter	Single Pack (BEFORE)	Dual Pack (AFTER)	Change
Energy available	158.4 Wh (1 × 158.4)	**316.8 Wh (2 × 158.4)**	**×2.0 (doubled)**
Nominal voltage	36V	**36V (unchanged)**	**= same**
Effective internal resistance	R_pack	**R_pack / 2 (halved)**	**÷2 (halved)**
Voltage sag under 8A load	4–6V drop observed (36V→~31V, STM32 cutoff)	**Eliminated (ΔV_sag ÷ 2)**	**■ Fixed**
Light runtime (2.0A, 72W)	2.20h (132 min)	**4.40h (264 min)**	**×2.0**
Normal runtime (3.5A, 126W)	1.26h (76 min)	**2.51h (151 min)**	**×2.0**
Heavy runtime (8.0A, 288W)	0.55h (33 min)	**1.10h (66 min)**	**×2.0**
Charge time (2×2A)	2.4h (single pack)	**2.4h (2×2A dual)**	**= same time**
Charge time (2×4A fast)	1.2h (single pack)	**1.2h (2×4A dual)**	**= same time**
System stability	BAT_DEAD cutoffs during combined turns	**Zero cutoffs observed in all manoeuvres**	**■ Stable**
Production viability	■ 76 min normal use not viable for daily living	**■ 151 min normal use exceeds daily assistive threshold**	**Viable**

Production Argument — Use in Slide Tagline

"A 76-minute normal-use runtime is insufficient for a patient's daily living needs. At 151 minutes on the dual-pack, RAFEEQ exceeds the threshold for a viable daily-use assistive device — while keeping charging time identical at 2.4 hours with dual standard chargers." (§4.3.2.4, Table 4.6)

How to Present This on ELEC-5

- Top of slide: the problem → fix story (2 cards, red problem / green solution)
- Centre: this full comparison table, highlighting the runtime rows with a colour sweep animation
- Bottom strip: the 3 key numbers large — 4.40h / 2.51h / 1.10h — with gold stat styling
- Tagline: "Runtime doubled. Sag eliminated. Dual charging. Same charge time." — gold italic

SECTION 8

Slide ELEC-6 — The 4-Layer Safety Net

Defence-in-depth. Each layer independent. Thesis §4.3.2.3 Figure 4.12.

ELEC-6	The 4-Layer Safety Net — Defence-in-Depth	DARK · 4 CARDS

Layout

Same 4-card grid pattern as CTRL-4. Dark glass cards. Staggered entrance. Each card: level number (ghost), icon, name, what it does, what it catches.

L1 — ■ Battery BMS · Hardware Layer · Latency: Hardware instantaneous

What it does	What it catches
VICTa pack integrated BMS. Enforces per-cell voltage bounds: 4.2V max / 2.5V min. Temperature cutoff at 60°C.	Over-charge, deep discharge, thermal runaway. Cannot be overridden by software — deepest protection layer.

L2 — ■ Inline Fuses · Hardware Layer · Latency: Microseconds

What it does	What it catches
Motor lines: 30A automotive blade fuses (one per positive line of Y-harness). Electronics pack: 10A master fuse upstream of all buck converters. 12AWG wiring throughout.	Short circuits. Wire overload. Sized per IEC 60364-5-52: I_derated=0.8×41A=32.8A → 30A fuse. Below derated limit, above 15A continuous rating — no nuisance tripping.

L3 — ■■ Software Watchdog · Firmware Layer · Latency: ≤500ms

What it does	What it catches
Arduino firmware monitors interval between consecutive valid V,W commands from RPi5. Threshold: 500ms.	RPi5 crash, USB disconnect, software freeze. On timeout: Arduino sends cmd=(0,0) to STM32 → motors freewheel. (§4.3.2.3, [43])

L4 — ■ Hardware E-Stop · Physical Layer · Latency: <1ms

What it does	What it catches
Latching mushroom-head button (normally-closed, NC) wired to Arduino hardware interrupt pin. ISR executes at highest CPU priority.	Any emergency. Bypasses all software. Writes cmd=(0,0) to STM32 in under 1ms — faster than any scheduler-based response. Complies with ISO 13482 redundant-stop requirements.

Key slide tagline (bottom, centred, gold italic): "Each layer is fully independent — failure of any one layer does not disable the others." (Direct quote from Figure 4.12 caption, thesis §4.3.2.3)

Card animation: stagger L1 at 380ms, L2 at 510ms, L3 at 640ms, L4 at 770ms. translateY(18px)→0, opacity 0→1, 0.5s cubic-bezier(0.16,1,0.3,1). Same as CTRL-4.

SECTION 9

Asset Reference — Images Required

Images Needed for Electrical Slides

File Name	Used In	Description	Status
STM32_FOC_Motor_Driver.png	ELEC-4 · driver board visual	YOUR actual STM32 board (same image as control slides). Already have.	✓ Have it
bldc_hub_motor.jpg	ELEC-4 · motor image	Your actual 8-inch BLDC hub motor as mounted on the wheelchair. Photograph it from the side clearly showing the wheel.	Photograph
battery_victa.jpg	ELEC-5 · battery visual	VICTa pack with XT60 connector. Photograph your actual pack.	Photograph
xt60_yharness.jpg	ELEC-5 · parallel fix visual	XT60 Y-harness connecting two packs. Photograph the actual hardware.	Photograph
estop_button.jpg	ELEC-6 · safety visual	Red mushroom E-stop button on wheelchair. Photograph it.	Photograph
fuse_holder.jpg	ELEC-6 · fuse visual	30A automotive blade fuse in holder on motor line. Photograph it.	Photograph
wiring_diagram.png	ELEC-3 · block diagram	Figure 4.10/4.11 from thesis — the full wiring diagram. Export from thesis PDF.	Extract from thesis

No video assets needed for electrical slides. All 6 slides are HTML/CSS/JS only.

SECTION 10

Presenter Speaking Notes — Electrical Section

Mohamed Morshedy presents ELEC-1 through ELEC-6 (health monitoring + IoT is his domain). These are the key verbatim points from thesis §4.3.2.

ELEC-1 — The Hook (~30s)

- "We have just seen the mechanical chassis. Right now it is inert hardware. To make it intelligent, we need an electrical backbone built around one principle: Total Power Isolation."
- "The reason for isolation is physics: our motor driver switches at 20kHz and can spike to 15 amps per phase. That switching noise would corrupt every serial packet and sensor reading if the computers shared the same battery. So we separated them completely."

ELEC-3 — Power Distribution (~1 min)

- "We run on two completely independent battery domains. The 36V traction pack — a VICTa 10S2P Samsung lithium-ion with 158Wh — powers only the motors through the STM32 driver."
- "The 12V electronics pack feeds three galvanically isolated DC–DC buck rails. Rail 1 is an XL4016 stepping to 5.1V at 5A for the Raspberry Pi 5 — we chose XL4016 over the cheaper LM2596 because the RPi5 draws up to 5A under heavy SLAM and inference load, and LM2596 would cause voltage warnings and CPU throttling."
- "Rails 2 and 3 are LM2596 converters at 5V/3A and 5V/1A for the peripherals and Arduino respectively. Each rail is independently fused."

ELEC-4 — Motors (~1 min)

- "Rafeeq is driven by two 350-watt brushless DC hub motors with 10-inch pneumatic tyres. We run them with sinusoidal Field Oriented Control at 20kHz — entirely inside the STM32."
- "Why FOC? Trapezoidal commutation gives you torque ripple — you feel it as vibration. For a seated patient, that is unacceptable. FOC maintains the stator current vector orthogonal to rotor flux at all times. The result is a completely smooth, silent ride."
- "Current draw changes with the terrain: 2 amps cruising slowly, 3.5 amps in typical mixed use, and up to 8 amps climbing a ramp. These numbers come from thesis Table 4.4."

ELEC-5 — Battery Runtime (~1.5 min)

- "During seated integration testing we observed a 4 to 6 volt sag on the 36-volt bus during combined forward-plus-rotation manoeuvres. The STM32 was triggering its BAT_DEAD protection and stopping the wheelchair mid-turn."
- "The primary fix was in firmware — we reduced DEFAULT_RATE from 800 to 300 to cut the current transient. As a secondary hardware fix, we connected two battery packs in parallel via an XT60 Y-harness. This halves the effective internal resistance and doubles the total energy from 158 to 316 watt-hours."
- "The result: normal-use runtime went from 76 minutes on a single pack to 151 minutes on dual packs. That is the difference between a device that is a prototype and one that is viable for daily living."
- "Charging time stayed the same — 2.4 hours with two standard 2-amp chargers, one per pack. 1.2 hours with two 4-amp fast chargers."

ELEC-6 — Safety (~1 min)

- "Because a human life depends on this machine, we implemented four completely independent safety layers. Any single layer failing does not disable the others."
- "Level 1 is the battery BMS — hardware that cannot be overridden by software. Level 2 is 30-amp automotive fuses sized per IEC 60364 — they physically break the circuit. Level 3 is a 500-millisecond software watchdog — if the Raspberry Pi crashes or the USB cable comes out, the Arduino detects the silence and sends zero commands to both motors within half a second."
- "Level 4 is the hardware emergency stop — a latching mushroom button wired directly to an Arduino interrupt pin. The ISR executes in under one millisecond. That is faster than any software scheduler can respond."

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